Machine Learning Project

Pediatric Appendicitis
Classification

Medical Relevance

- Common childhood disease requiring surgery
- Children rarely show classic symptoms

False Negatives:
Risk of perforation

False Positives:
Risk of surgery

Dataset

Hospital St. Hedwig, https://www.barmherzige-hedwig.de/unser-haus.html

Regensburg pediatric appendicitis dataset

782 → 774 patients (missingness)

53 → 82 features (one-hot encoding, engineered features)

- Median imputation for numeric features
- Mode/Sentinel imputation for binary features
- Add indicator for observed features to keep information of missing values

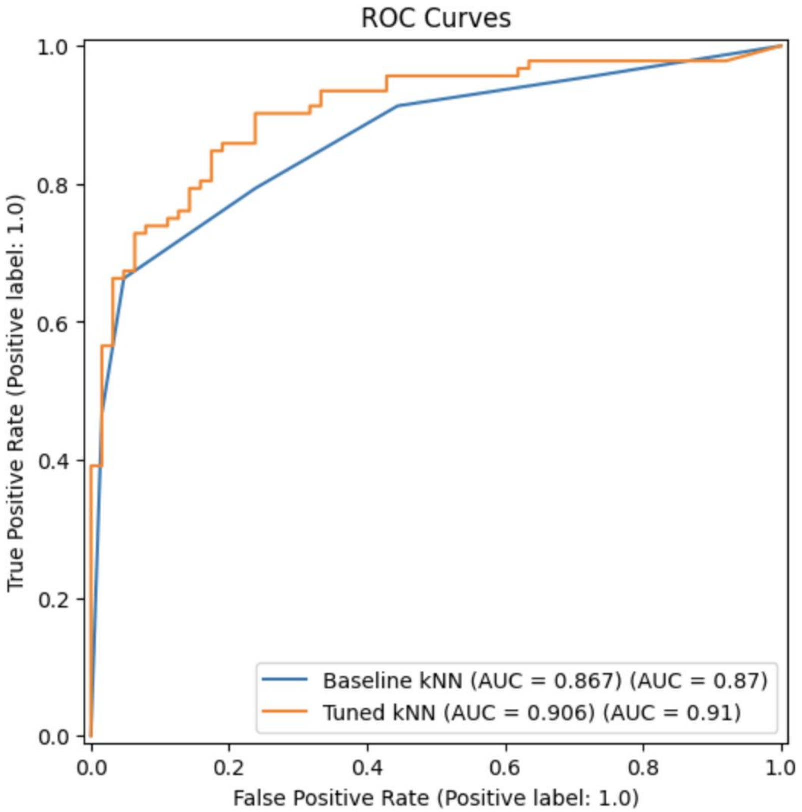
The Models Used

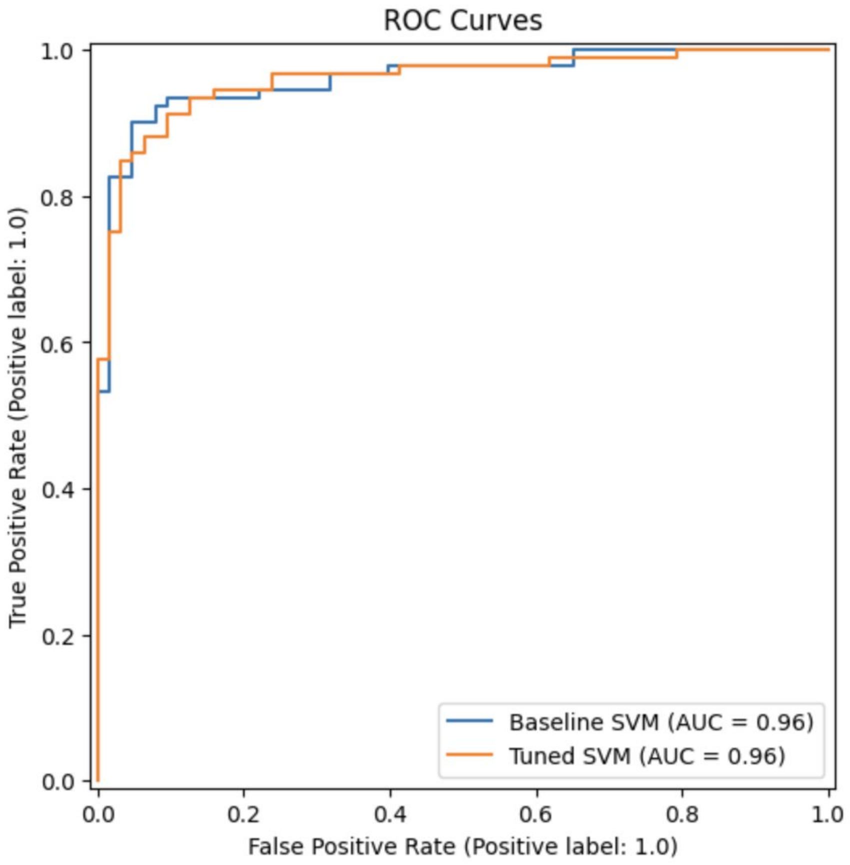
- Six classification algorithms
- Baseline vs. tuned models
- Hyperparameter tuning with GridSearchCV
- Evaluation: Accuracy, Precision, Recall, F1, ROC-AUC

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Baseline kNN	0.781	0.830	0.793	0.811	0.867
Tuned kNN	0.826	0.874	0.826	0.849	0.906
PCA + Tuned kNN	0.800	0.802	0.880	0.839	0.907
Baseline SVM	0.916	0.934	0.924	0.929	0.961
Tuned SVM	0.897	0.913	0.913	0.913	0.961
Decision Tree (baseline)	0.910	0.906	0.946	0.926	0.901
Decision Tree (tuned)	0.916	0.916	0.946	0.930	0.900
Random Forest (baseline)	0.955	0.957	0.967	0.962	0.969
Random Forest (tuned)	0.955	0.957	0.967	0.962	0.979
XGBoost (baseline)	0.955	0.947	0.978	0.963	0.977
XGBoost (tuned)	0.961	0.957	0.978	0.968	0.980
LightGBM (baseline)	0.955	0.957	0.967	0.962	0.968
LightGBM (tuned)	0.948	0.957	0.957	0.957	0.966

Table 1: Comparison of all explored models predicting the primary target *Diagnosis*

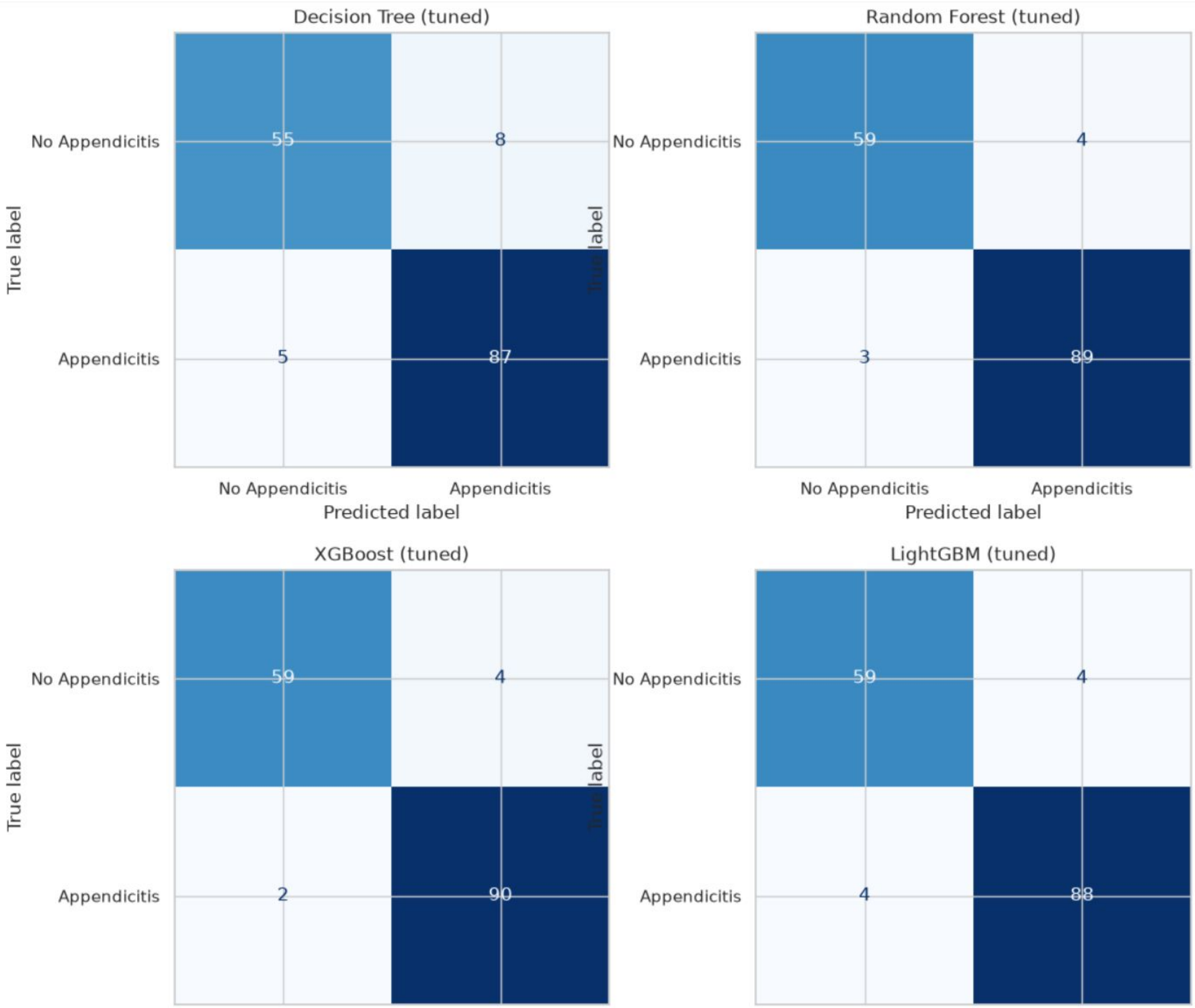
kNN and SVM ROC Curves

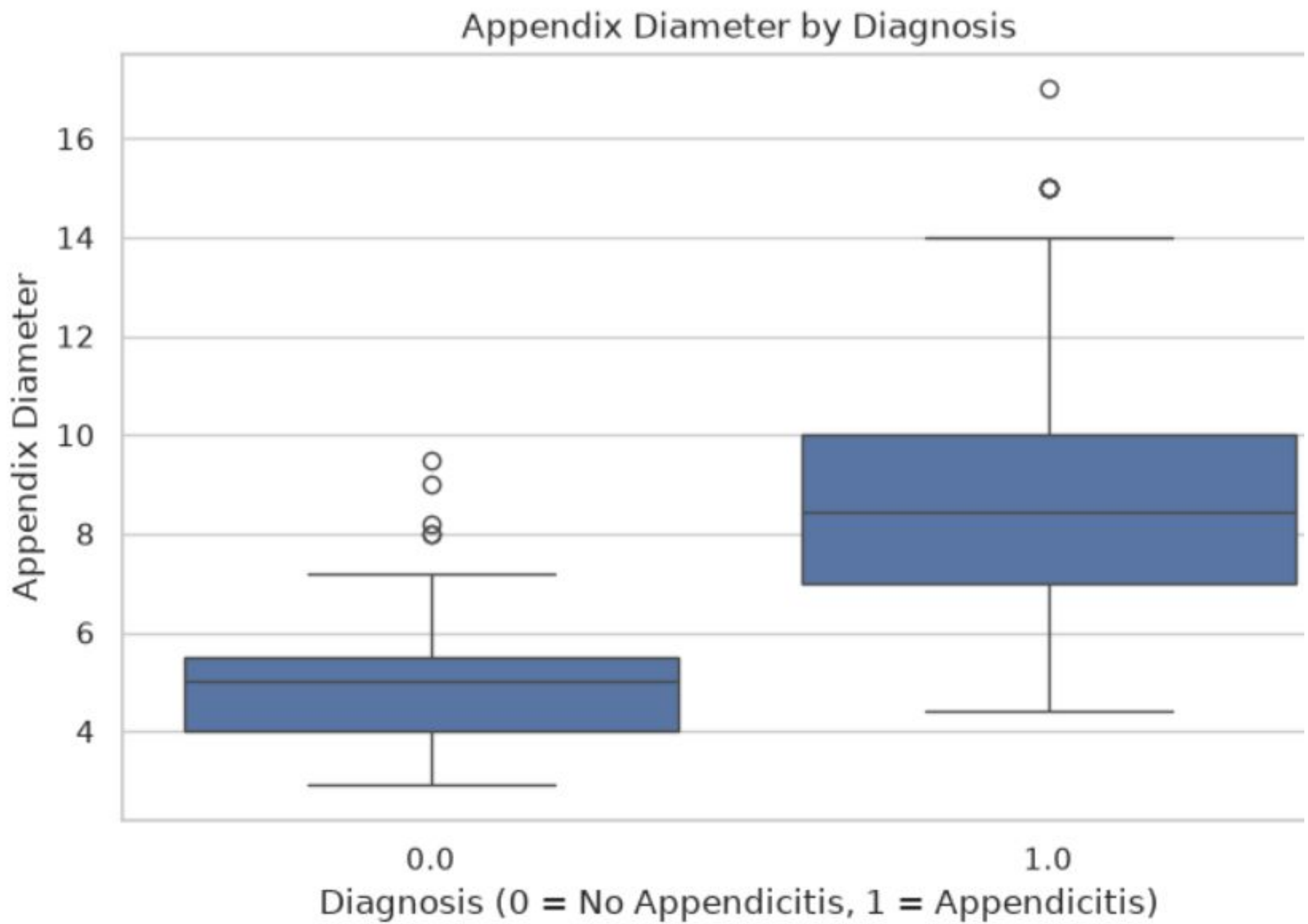




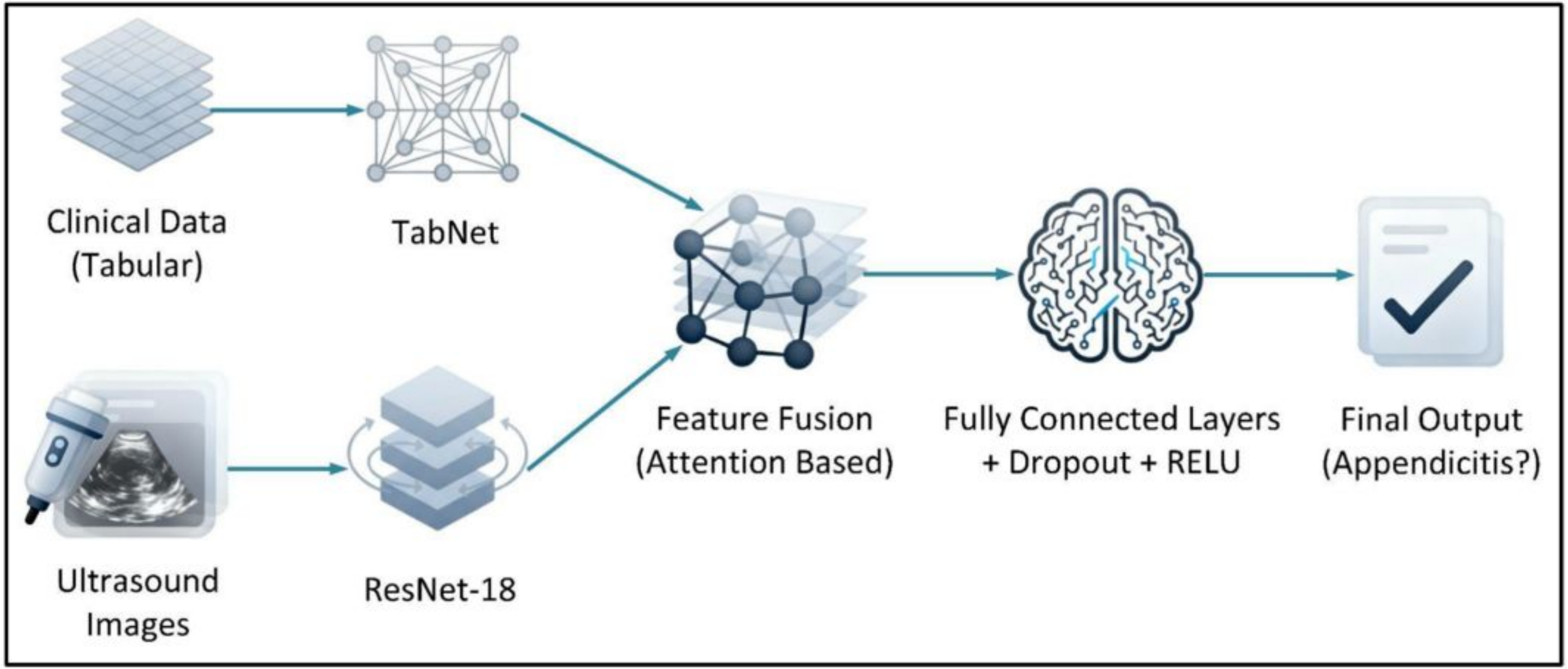
XGBoost:

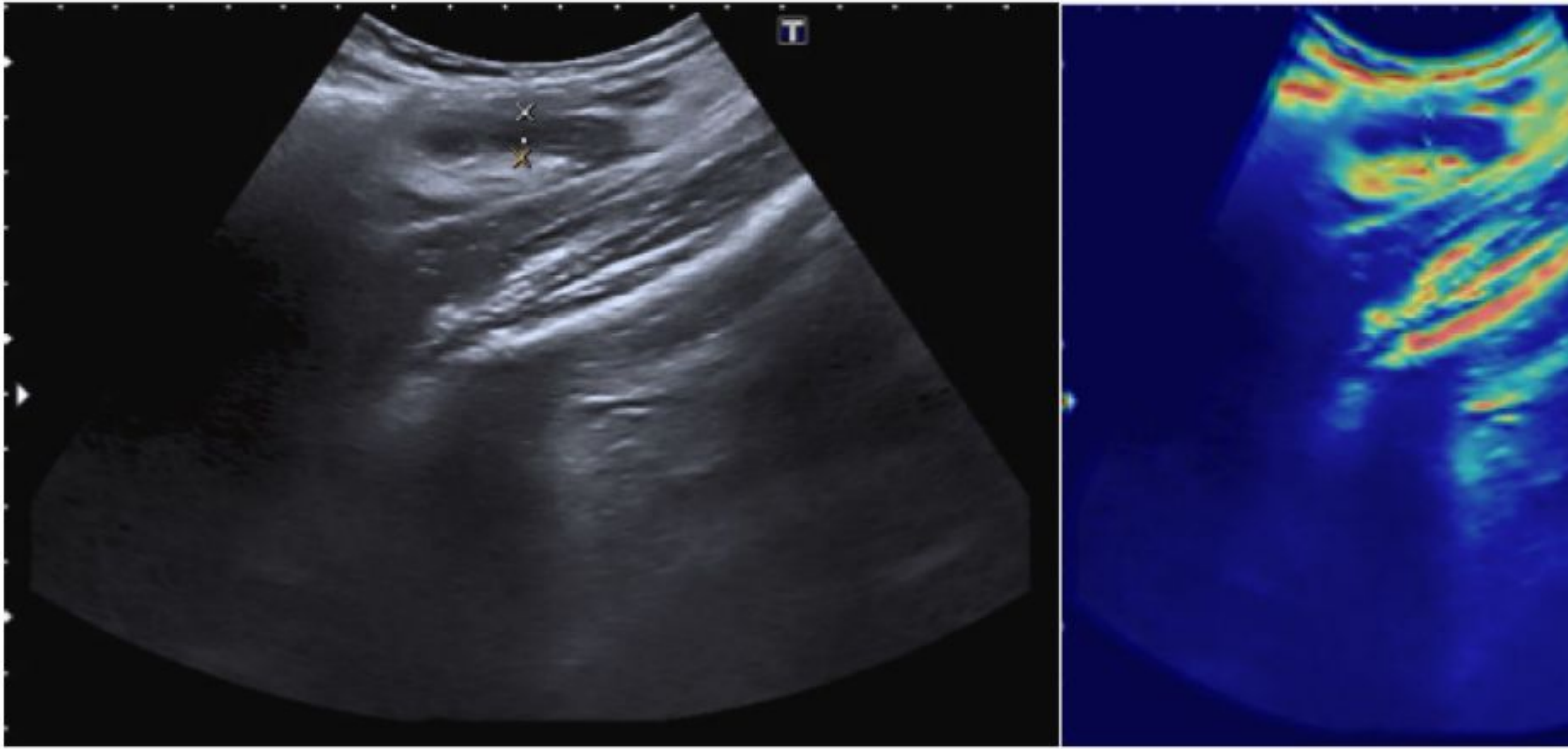
- Accuracy of 96% on test data
- ROC-AUC: 0.98





Appendix Diameter by Diagnosis
Appendix Diameter
16
14
12
10
8
6
4
0.0
1.0
Diagnosis (0 = No Appendicitis, 1 = Appendicitis)


Clinical Data
(Tabular)
TabNet
Ultrasound
Images
ResNet-18
Feature Fusion
(Attention Based)
Fully Connected Layers
+ Dropout + RELU
Final Output
(Appendicitis?)



Thanks to the original authors, especially for collecting such valuable diagnostic data!

Here are our sources, as it is important to give credit to the scientific work of others:

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